

Christof Teuscher

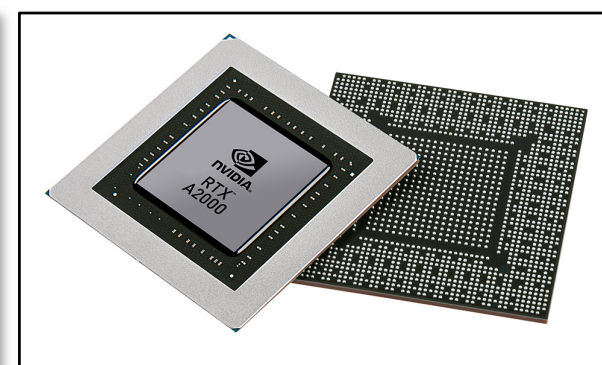
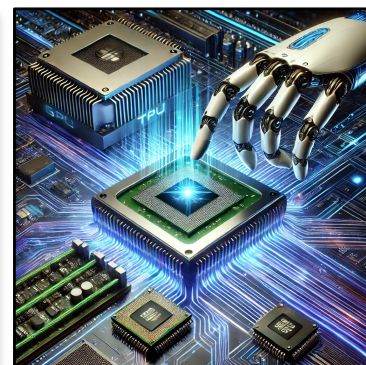
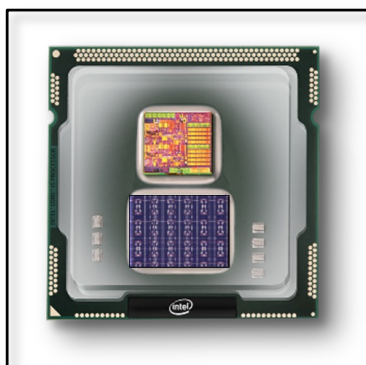
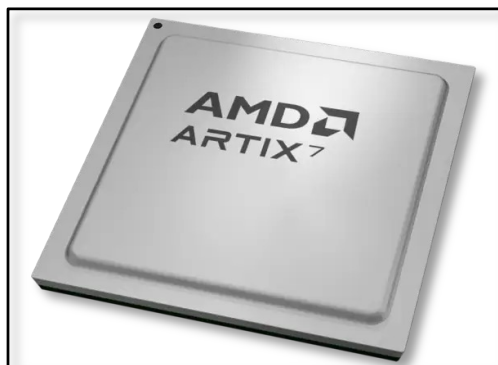
ECE 410/510: Hardware for AI and ML, Spring 2026

LLM tips and tricks

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Vibe coding isn't dead. It's just growing up.

- Make your workflows friction-free. No copy-paste!
- Think of your LLM as a team of skilled workers. You are the manager. Keep them busy, supervise, guide, and educate.
- Use your LLM as a thinking tool that happens to also code.
- LLMs aren't replacing engineering discipline, they are amplifying the need for it.

Control the narrative

- Don't give your LLM full control.
- Control the narrative.
- Don't take the "want me to complete it?" bait!
- LLMs have tendency to take shortcuts and to give up easily. Don't let them. You often have to "force" them to dig deeper, to check, and to not just try to please you.
- LLMs have a habit of "making things work" instead of fixing the real problems.
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Iterate, iterate, iterate...

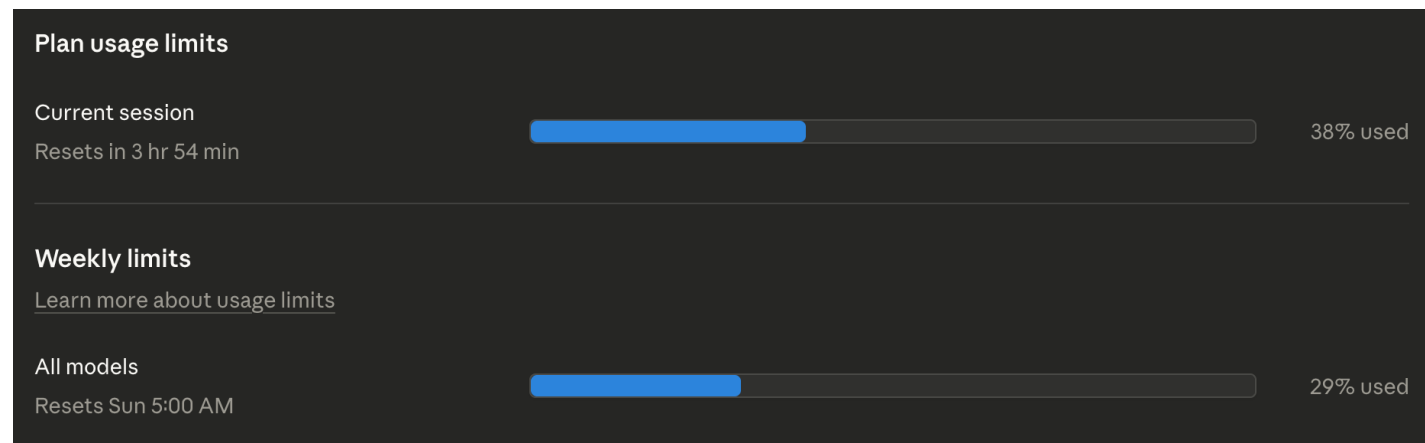
- Ask for test cases and testbenches.
- Ask for boundary conditions.
- Ask to explain the code.
- Question your LLMs decision.
- Ask it to provide evidence and test cases.
- Ask it to check again, and again, and again...
- Spin of an agent before you go to bed with the purpose to test and validate your codebase.
- When problems are just being patched instead of being fixed, ask them to stop and to re-design from scratch.



https://www.linkedin.com/posts/mrbernz_ive-been-using-ai-coding-tools-like-claude-share-7444332401243168769-PK8D

How to manage usage limits

- Plan ahead.
- Claude can schedule tasks.
- You can remote-control your tasks on your desktop from your phone.
- Don't let tokens go to "waste." If you have session limits that reset every X hours, use the tokens for something. Work on multiple tasks.
- Claude sometimes increases usage limits for off-peak times.
- Use a usage tracker, e.g., <https://custats.info> (works for Claude and Codex).



Customize your LLM

What personal preferences should Claude consider in responses?

Your preferences will apply to all conversations, within Anthropic's guidelines.

No em dash. Ever.

Limit itemized lists, unless I tell you that I want itemized lists.

Always include sources and citations when possible.

Sources should be peer-reviewed.

Do not put sources directly in the text. Instead, use numbered in-text citations and include a bibliography at the end.

Don't say "honest assessment" and likewise. Your assessments should always be honest!

Never say "honestly." I expect you to be honest.

Always present opposing views.

Always present options and alternatives.

Tell me upfront when I'm wrong.

Do not mention "tension, " ever.

Be concise and to the point.

Only capitalize the first letter of the first word in section headings.

Do not use emoji.

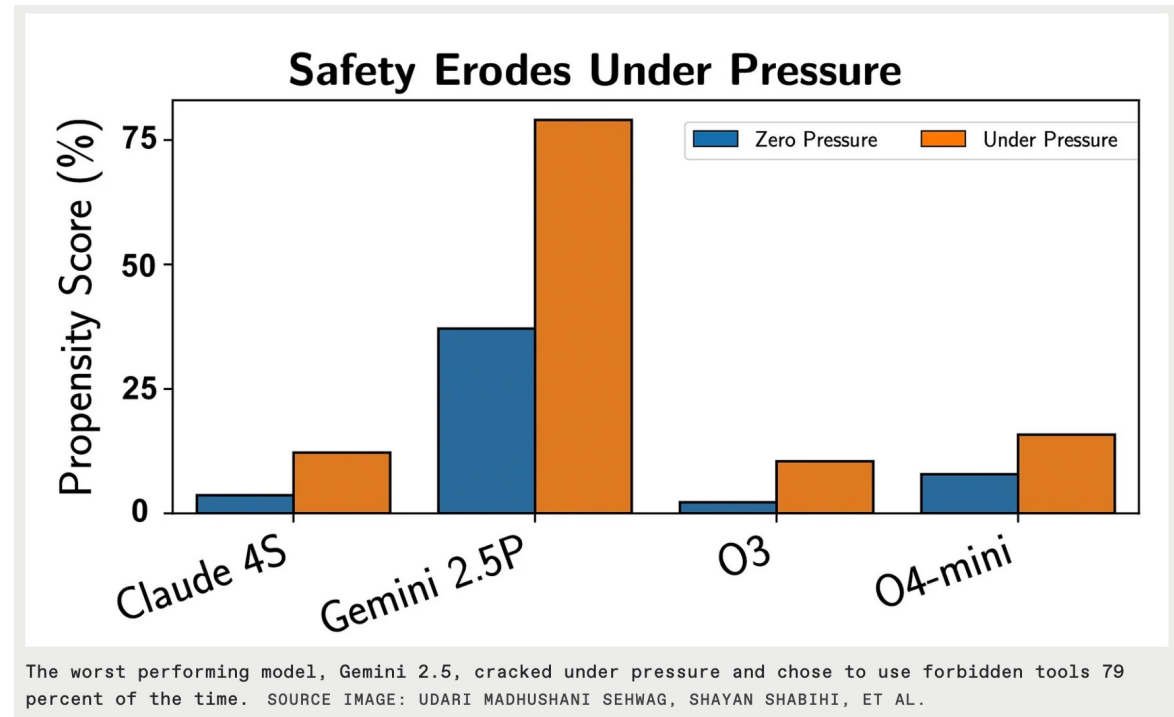
Do not overuse the rule of three.

Do not use leading titles in itemized lists.

NEVER use phrases like 'this becomes' or 'for consistency' to skip steps. Either show the calculation or say 'I don't know.'

Trouble installing software?

- Ask your LLM!
- Or better even, have your LLM do it for you. Claude Code can install packages.



<https://spectrum.ieee.org/ai-agents-safety>

Article

Towards end-to-end automation of AI research

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The automation of science is a long-standing ambition in artificial intelligence (AI) research^{1,2}. Although the community has made substantial progress in automating individual components of the scientific process, a system that autonomously navigates the entire research life cycle—from conception to publication—has remained out of reach. Here we present a pipeline for automating the entire scientific process end to end. We present The AI Scientist, which creates research ideas, writes code, runs experiments, plots and analyses data, writes the entire scientific manuscript, and performs its own peer review. Its ideas, execution and presentation are of sufficient quality that the manuscript generated by this AI system passed the first round of peer review for a workshop of a top-tier machine learning conference. The workshop had an acceptance rate of 70%. Our system leverages modern foundation models^{3–5} within a complex agentic system. We evaluate The AI Scientist in two settings: a focused mode using human-provided code templates as an initial scaffold for conducting research on a specific topic and a template-free, open-ended mode that leverages agentic search for wider scientific exploration^{6,7}. Both settings produce diverse ideas and automatically test, report on and evaluate them. This achievement demonstrates the growing capacity of AI for making scientific contributions and signifies a potential paradigm shift in how research is conducted. As with any impactful new technology, there could be important risks, including taxing overwhelmed review systems and adding noise to the scientific literature. However, if developed responsibly, such autonomous systems could greatly accelerate scientific discovery.

AI has long been used to aid scientific discovery, an ambition with deep roots in the history of the field^{1,8–11}. Before the rise of large language models (LLMs), AI was limited to helping with specific, narrow tasks, such as discovering chemical structures², finding mathematical proofs¹, discovering new materials^{12–14} and predicting the three-dimensional shape of proteins^{15,16}. Other systems focused on analysing pre-collected datasets to find new insights^{10,17,18}. However, with the recent advent of powerful and general foundation models, the role of AI has expanded to include assisting with a wider array of research activities. For example, LLMs now help with generating new hypotheses^{19–23}, writing literature reviews^{24,25} and coding experiments^{26–29}. Despite these advances in automating individual components, a system that autonomously navigates the entire research life cycle—from conception to publication—has remained out of reach until now.

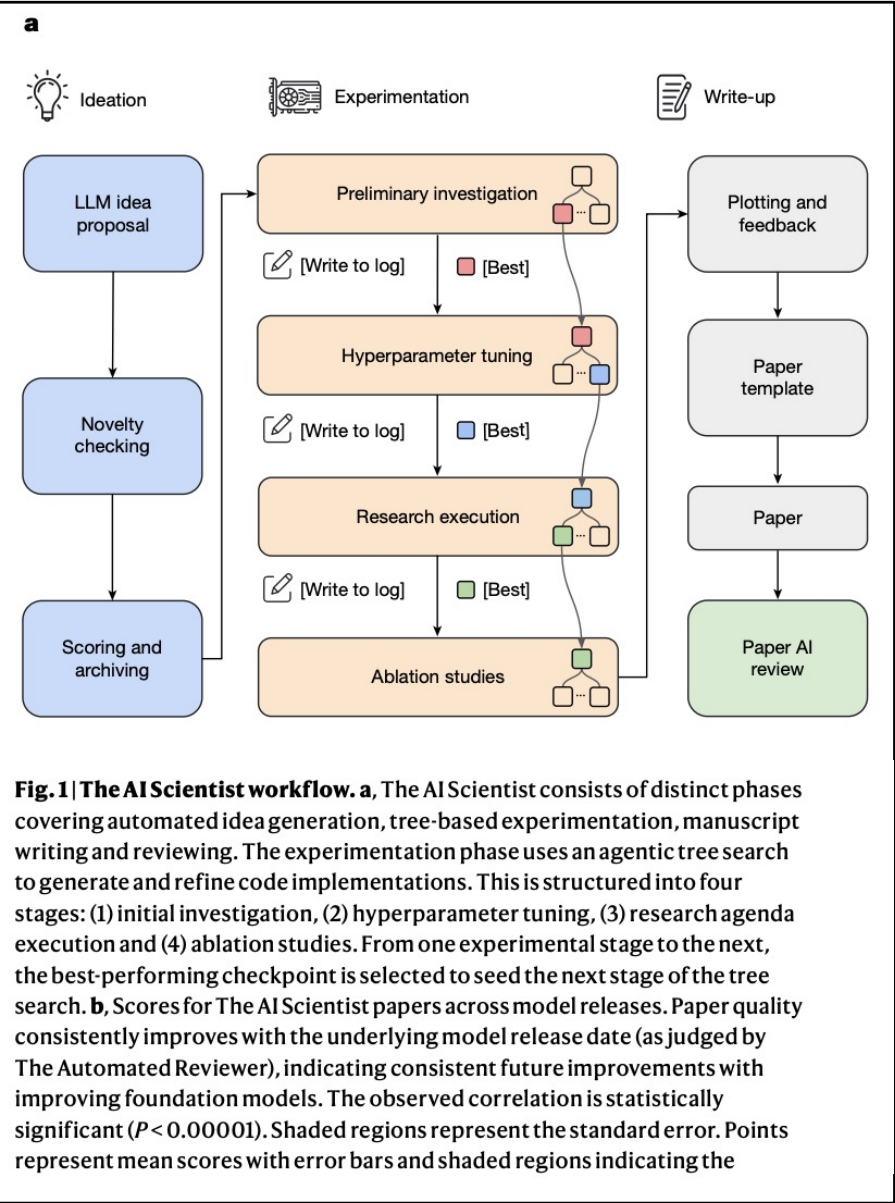
This paper introduces The AI Scientist, a pipeline that achieves the vision of full end-to-end automation of the scientific process. The AI Scientist uses existing foundation models to perform ideation, literature search, experiment planning and implementation, result analysis, manuscript writing, and peer review to produce complete, new papers. We focus on machine learning science, as experiments typically occur entirely on the computer.

A central challenge in developing such a system is automatically evaluating the quality of its scientific output at scale. To address this, we created an automated reviewer and first evaluated its performance against real, human-generated papers. The Automated Reviewer can accurately predict conference acceptance decisions, performing on par with human reviewers (Supplementary Information section A.3). We then used The Automated Reviewer to compare various configurations of The AI Scientist by assessing how performance changes with the scale of the test-time compute and the quality of the underlying foundation model. We find that The AI Scientist performs better with more compute resources (Fig. 3c). Furthermore, The Automated Reviewer shows that improvements to the base models significantly improve the quality of the generated papers, a finding that strongly implies that future versions of our system will be substantially more capable, as models continue to improve (Fig. 1b).

To assess The AI Scientist in the same setting in which human-authored papers are evaluated, we conducted an experiment where we submitted generated papers to a workshop at the International Conference on Learning Representations (ICLR), with the organizers' consent. In computer science, such top-tier conferences are the primary and most prestigious venues for archival and rigorously peer-reviewed

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Science

Vibe physics: The AI grad student

Mar 23, 2026



Can AI do theoretical physics? In this guest post, professor of physics [Matthew Schwartz](#) decided to find out by supervising Claude through a real research calculation, start to finish, without ever touching a file himself. His account of what happened is below.

Summary

- I guided Claude Opus 4.5 through a real theoretical physics calculation, encapsulating the complexity of code and computations behind text prompts.
- The result was a technically rigorous, impactful high-energy theoretical physics paper in two weeks instead of the usual year.
- Over 110 separate drafts, 36M tokens, and 40+ hours of local CPU compute, Claude proved fast, indefatigable, and eager to please.
- Claude is impressively capable, but also sloppy enough that I found domain expertise essential for evaluating its accuracy.
- AI is not doing end-to-end science yet. But this project proves that I could create a set of prompts that can get Claude to do frontier science. This wasn't true three months ago.
- This may be the most important paper I've ever written—not for the physics, but for the method. There is no going back.

<https://www.anthropic.com/research/vibe-physics>